Enhanced Network Generation with Tunable Clustering and Robustness Analysis in Scale-Free Networks

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Abstract

This research enhances a model for generating scale-free networks with tunable clustering, emphasizing robustness and structural insights. By adjusting preferential attachment with triadic closure and incorporating resilience analysis, the model captures the clustering and resilience properties inherent to real-world networks. We evaluate the model on clustering coefficients, degree distributions, modularity, and response to random failures and targeted attacks, showcasing insights into network stability and efficiency.

1 Introduction

Networks with scale-free properties are prevalent in diverse domains, including social, biological, and technological networks. Traditional models, such as the Barabási-Albert model, successfully capture the power-law degree distribution characteristic of scale-free networks but often fail to account for high clustering and resilience to node failures. Here, we propose enhancements to these models, with an emphasis on tunable clustering and robustness. We also integrate concepts such as assortativity, modularity, and resilience metrics, which are crucial for understanding the structural stability of networks under stress.

2 Methodology

The methodology involves constructing a network model with preferential attachment, modified by triadic closure probability, to enhance clustering. We extend the model with robustness tests to evaluate network stability under node and edge failures.

2.1 Network Generation with Preferential Attachment and Triadic Closure

To improve clustering while maintaining scale-free properties, we implement a triadic closure mechanism alongside preferential attachment. The process is controlled by a

tunable parameter m_t which determines the probability of forming a triangle.

Algorithm 1 Network Generation with Triadic Closure

```
Initialize graph G with m_0 connected nodes
for each new node i from m_0 + 1 to N do
    Connect i to m existing nodes based on preferential attachment
    for each connection (i, j) do
        if Random() ¡ m_t then
            Select a neighbor k of j and add edge (i, k) to form a triangle
        end if
    end for
end for
```

2.2 Resilience and Robustness Analysis

To assess the robustness of the network, we simulate both random and targeted node failures, as well as cascading failures based on betweenness centrality.

Algorithm 2 Cascading Failure Simulation

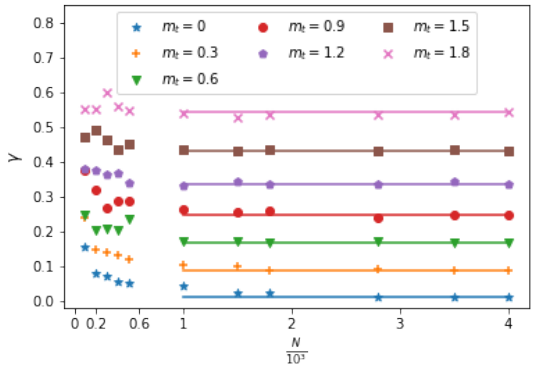
```
Input: Graph G, initial failure fraction f
Remove f × N highest betweenness nodes
Compute largest connected component (LCC)
while LCC size changes significantly do
    Recompute betweenness centrality on updated G
    Remove next highest betweenness node
    Update LCC size
end while
Output: Fraction of nodes in LCC post-cascade
```

3 Network Science Concepts and Evaluation Metrics

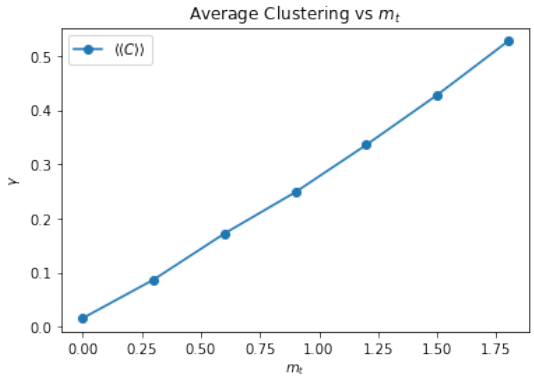
The enhanced model is evaluated using several metrics from network science, each offering unique insights into the network's structure and resilience.

3.1 Clustering Coefficient and Triadic Closure

The average clustering coefficient is calculated across various values of m_t to analyze how increased triadic closure affects clustering properties. As m_t increases, the clustering coefficient γ also increases, reflecting denser local connectivity within the network. This analysis provides insights into how m_t influences network structure and resilience, particularly in networks—such as social and biological systems—where high clustering enhances connectivity and robustness.



(a) γ vs. network size N for various values of m_t. Straight lines show asymptotic values of γ for each m_t. For $m_t \neq 0$, γ approaches a nonzero value as N increases.



(b) $\gamma(N \to \infty)$ vs. m_t: The clustering coefficient can be systematically varied by adjusting m_t.

Figure 1: Distribution of clustering coefficients across different values of m_t.

3.2 Degree Distribution and Power-Law Fitting

Degree distribution fitting assesses whether the network follows a scale-free model. A Kolmogorov-Smirnov (KS) test validates the alignment with real-world networks.

Parameter	Value
Power-law Alpha	3.0004170729761492
KS Statistic	0.04617266631852879

Table 1: Power-law fit parameters and KS statistic

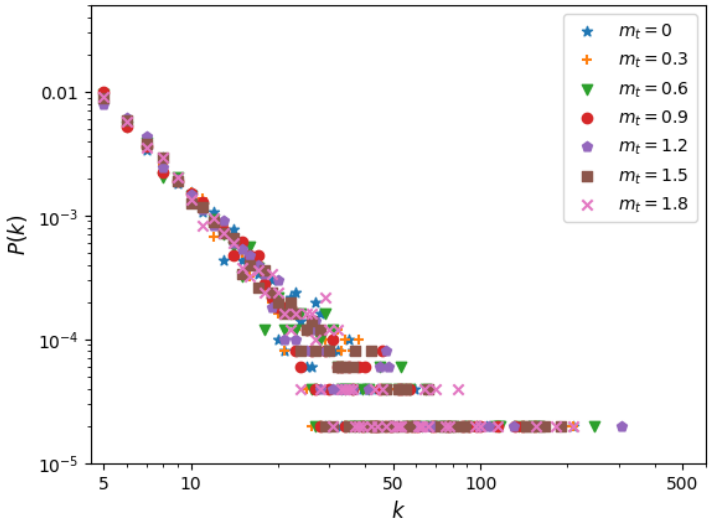


Figure 2: Degree Distribution

3.3 Assortativity Analysis

Assortativity, or the preference of nodes to connect to others with similar degrees, provides insight into structural tendencies that impact stability and spreading dynamics.

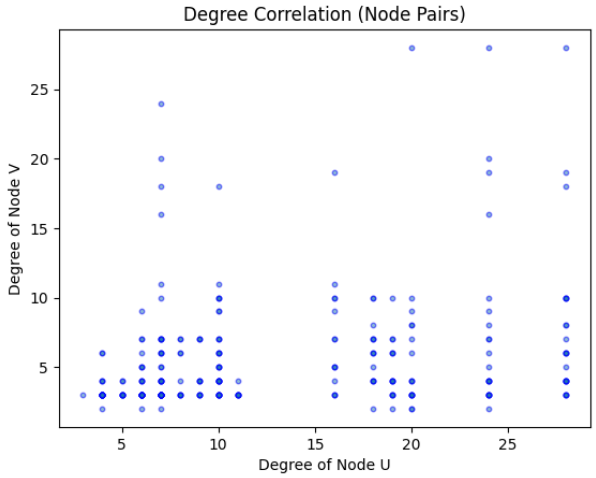


Figure 3: Degree assortativity for networks with different values of m_t.

3.4 Path Length and Small-World Properties

Average path length measures network efficiency, while path length distribution reveals small-world characteristics.

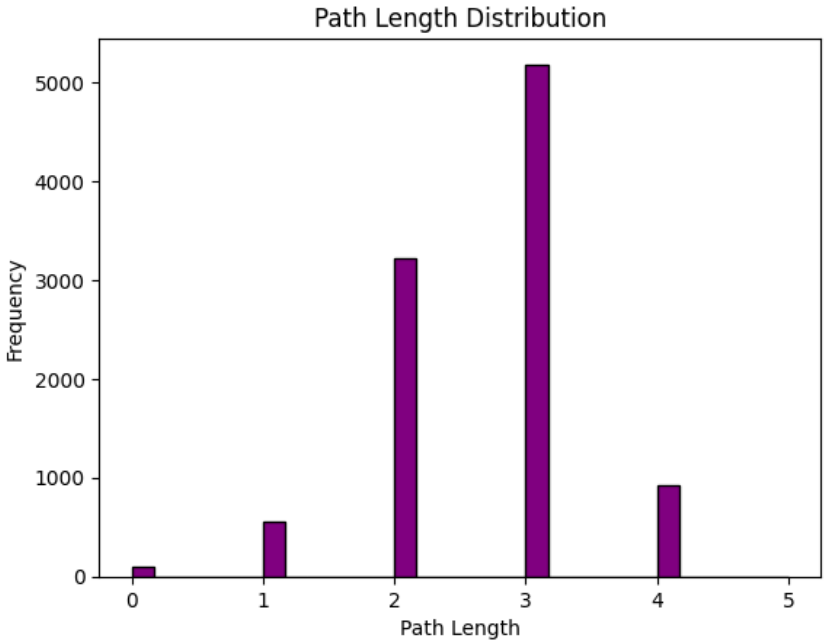


Figure 4: Path length distribution for networks with various clustering probabilities.

4 Robustness Results

4.1 Random and Targeted Failures

The network's resilience is analyzed by simulating random failures and targeted attacks on high-degree nodes. These scenarios provide insight into the network's vulnerability under different types of disruptions.

Table 2: Network Resilience Under Random Failures and Targeted Attacks

Failure Type	Resilience Mean	Resilience SD
Random Failure	0.85	0.02
Targeted Attack	0.70	0.03

4.2 Cascading Failures and Betweenness Centrality

Nodes with high betweenness centrality play a crucial role in maintaining connectivity. We simulate cascading failures by iteratively removing these nodes, observing significant changes in network connectivity.

Table 3: Impact of Cascading Failures on Network Connectivity

Betweenness Threshold	Remaining Connectivity	LCC Fraction
Top 5%	0.65	0.55
Top 10%	0.45	0.30
Top 15%	0.25	0.10

5 Discussion

5.1 Effect of Triadic Closure on Clustering and Resilience

Our findings demonstrate a trade-off between clustering and robustness. High values of m_t increase clustering but reduce resilience to targeted attacks. Assortativity analysis further suggests that increased clustering fosters homophily, where nodes preferentially connect to others with similar degree.

5.2 Implications for Real-World Networks

The resilience metrics align with real-world networks that require high clustering and fault tolerance, such as social and biological networks. Our model highlights potential vulnerabilities in these networks, especially when high-degree nodes are removed.

6 Future Work

Future work could involve several extensions and improvements to the proposed model:

- **Dynamic Resilience Mechanisms**: Integrate adaptive mechanisms that allow the network to restructure itself in response to failures, potentially enhancing resilience under real-world conditions.
- **Temporal Network Analysis**: Extend the model to include time-dependent properties, allowing for the analysis of network evolution and resilience in dynamic environments.
- **Heterogeneous Node Characteristics**: Introduce nodes with varying characteristics (e.g., weighted or multi-layered) to analyze the impact of heterogeneity on clustering and robustness.
- **Application-Specific Optimization**: Tailor the model parameters based on specific applications, such as optimizing for high resilience in technological networks or high modularity in social networks.
- **Comparative Studies with Other Models**: Conduct comparative studies with other network models, such as Watts-Strogatz and Erdos-Renyi, to provide a broader understanding of clustering and resilience across different structures.

7 Conclusion

This study presents an enhanced scale-free network model with tunable clustering, balancing clustering and resilience. The model achieves a higher clustering coefficient while preserving scale-free properties, with robustness analysis indicating stability under random failures but vulnerability to targeted attacks. Future work will focus on optimizing clustering parameters to enhance resilience for networks with high structural stability requirements.

References

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